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DEPRESSION OF CEREBRAL GLUCOSE UTILIZATION DURING ANIMAL HYPNOSIS IN THE RABBIT

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The rates of local cerebral glucose utilization have been measured in normal conscious and hypnotized rabbits by the [^{14}C]deoxyglucose method. In control rabbits the rates vary widely throughout the brain, with the values in gray matter broadly distributed around an average which is about 3 times greater than that of white matter. The higher values are in structures of the auditory system (superior olive, inferior colliculus, auditory cortex). Animal hypnosis reduces the rates of glucose utilization in all structures of the rabbit brain, particularly in the caudate nucleus, putamen and sensory and motor cortices.

Animal hypnosis (tonic immobility) in the rabbit is characterized by immobility, depression of spinal reflexes [1] and suppression of behavioral and EEG responses to prolonged noxious stimuli [4–6]. During hypnosis blood pressure and heart rate do not differ from control values, although a series of oscillations occur at the time of the manipulation to produce immobility [3, 8]. After a single induction of hypnosis testosterone metabolism at hypothalamic levels is greatly reduced [7].

The [^{14}C]deoxyglucose method [16] provides a mean to measure the rates of local cerebral glucose utilization (LCGU) simultaneously in all the macroscopic structures of the brain. Because functional activity and energy metabolism appear to be closely related in the nervous system, local alterations in glucose utilization accompany and reflect local changes in functional activities in the brain. In the present study, we utilized this method to ascertain what changes occur in the metabolic rates of different regions of the CNS during animal hypnosis.

The experiments were performed in 10 adult rabbits (New Zealand White) weighing approximately 2.0–2.6 kg. Each rabbit was prepared for the experiment by the insertion, under local anesthesia, of polyethylene catheters into auricular artery and vein. Approximately 1 h later, each rabbit was given a pulse of [^{14}C]deoxyglucose (50 μCi/kg in 0.5 ml saline) through the venous catheter. Timed blood samples were drawn from the artery during the subsequent 42–43 min for the

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determination of the arterial plasma [^{14}C]deoxyglucose and glucose concentration curves [16]. Plasma glucose concentration was measured with enzymatic techniques by means of a Beckman spectrometer (Beckman Instruments) and plasma [^{14}C]deoxyglucose concentration was determined in a liquid scintillation counter (Packard Instruments). At the end of the procedure animals were sacrificed by intravenous injection of barbiturates mixed with KCl, and their brains were rapidly removed and frozen in liquid nitrogen. The frozen brains were cut into 20 μm sections in a Cryocut microtome (American Optical) and one section every 150 μm was mounted on a slide and subjected to autoradiography. The LCGUs of various CNS structures, except the cerebellum, were determined from the measured optical density for each CNS structure on the autoradiogram and from plasma glucose, and [^{14}C]deoxyglucose concentration by means of an operational equation as previously described by Sokoloff et al. [17]. Optical densities were measured by means of a Photovolt densitometer (Photovolt Corp.).

The experiments were performed in 5 sessions. On each experimental session two identically prepared rabbits were used, one for hypnosis and one for control. Both animals were cage acclimatized and had never been previously submitted to hypnosis induction. Hypnosis was produced by placing the animal on its back in a wooden trough [9] and maintaining it there until fully relaxed. During hypnosis the animal was not exposed to sudden, external stimuli and termination occurred spontaneously. The first movement during hypnosis monitored the end of the immobility; the animal was removed from the trough and another hypnosis induction immediately followed. A series of hypnosis episodes was produced for 42–43 min. In the control animal, hypnosis inductions of equal durations and at the same latencies from the tracer injection as the hypnosis animal were produced. However, in this control animal, the manipulations of the induction were performed in such a manner as to prevent the occurrence of the immobility response. During intervals between manipulations the control animal remained unrestrained in a box ($25 \times 40 \times 50$ cm) open on the top. For the entire experimental period white noise attenuated environmental acoustical stimuli.

As shown in Fig. 1, the number and the duration of hypnosis episodes varied over a wide range. However, the duration of different experimental periods (hypnosis induction, hypnosis, procedures between the end of one hypnotic episode and the beginning of the following one) was approximately the same in all the animals (Table I). In the rabbits submitted to hypnosis treatment, cerebral glucose utilization decreased in all the structures studied. Table II shows that the greatest depressions occurred in the cerebral cortex (sensory, visual, and motor cortex) and in several subcortical structures (caudate-putamen, amygdala, septal nucleus, hippocampus and thalamus). It has to be emphasized that the auditory structures, which displayed a high rate of glucose utilization in control animals, were less affected than the structures involved in motor control (motor cortex, caudate-putamen).

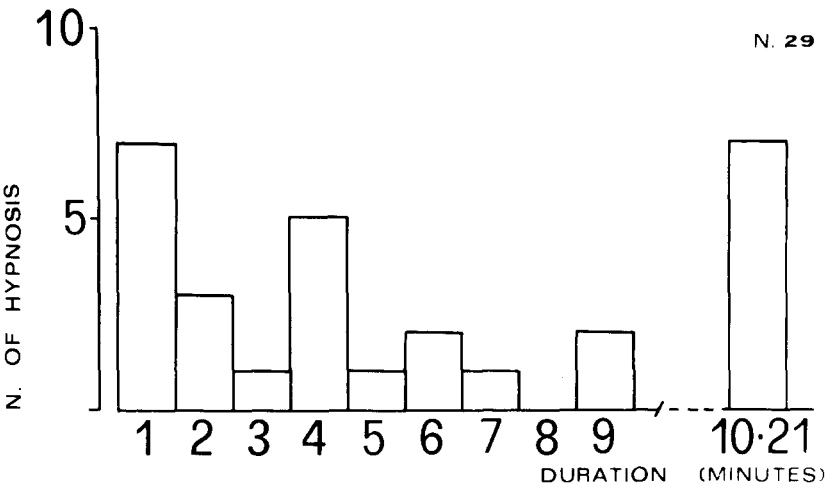


Fig. 1. Histogram of the duration of hypnosis episodes (5 rabbits, 29 episodes).

TABLE I

DURATIONS (MINUTES) OF THE EXPERIMENTAL CONDITIONS

Hypnosis, induction and interval procedure durations are the times (minutes) spent in each condition over the total experiment duration; % hypnosis is the percentage of total experimental time spent in hypnosis immobility.

Expt. No.	Number of hypnosis episodes	Hypnosis duration (min)	Induction duration (min)	Interval procedure duration (min)	Total experiment duration (min)	% Hypnosis (min)
1	3	36.33	1.66	4.51	42.50	85.4
3	7	36.25	2.25	4.33	42.83	84.6
5	4	37.33	2.75	2.67	42.75	87.3
7	6	30.66	4.50	6.41	41.57	73.7
9	9	35.25	3.16	3.31	41.72	84.4
Mean		35.16	2.86	4.24	42.47	83.1

Overall depression of cerebral glucose utilization has been reported following deep anesthesia [13, 16] and injection of β-endorphin into periaqueductal gray matter [11], whereas a generalized increase occurs following motor activity (swimming) [14] and electroanesthesia by extracranial electrodes [15]. In the present study the amount of motor activity performed by the hypnosis animal did not differ from the control one since, for the latter, limited room for movements was available. It is unlikely that the decrease of glucose utilization might be related to postural changes occurring during animal hypnosis since it is well known that, at

TABLE II

LOCAL CEREBRAL GLUCOSE UTILIZATION (μmol/100 g/min)

Values are means ± S.E. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.005$ by Student's t-test.

Structure	Control (n = 5)	Hypnosis (n = 5)	% Decrease
Visual cortex	76.4 ± 8.8	50.3 ± 7.1	34.1**
Auditory cortex	102.3 ± 9.6	80.1 ± 9.8	21.7*
Parietal cortex	73.1 ± 7.8	55.7 ± 3.4	23.8*
Sensory cortex	75.4 ± 7.5	45.6 ± 6.8	39.5***
Motor cortex	89.9 ± 6.2	63.2 ± 6.2	29.7***
Thalamus	80.1 ± 5.9	61.6 ± 1.9	23.1***
Medial geniculate body	108.2 ± 1.9	94.7 ± 5.5	12.5*
Lateral geniculate body	80.7 ± 4.9	67.2 ± 4.0	16.7
Hypothalamus	62.0 ± 3.3	52.2 ± 4.8	15.8**
Mammillary body	67.6 ± 2.8	58.1 ± 2.4	14.0**
Hippocampus	58.1 ± 2.5	44.5 ± 4.1	23.4***
Amygdala	39.1 ± 1.4	28.2 ± 2.7	27.9**
Caudate-putamen	85.8 ± 5.2	59.4 ± 7.1	30.5***
Globus pallidus	42.1 ± 2.3	34.5 ± 2.7	18.0*
Septal nucleus	48.3 ± 1.9	35.4 ± 1.8	26.7***
Superior olive	94.6 ± 4.9	85.1 ± 8.3	10.0
Substantia nigra	48.3 ± 3.2	40.0 ± 3.9	17.2
Inferior colliculus	125.5 ± 3.2	113.2 ± 3.6	9.8
Superior colliculus	62.1 ± 1.4	56.8 ± 2.0	8.5
Pontine gray matter	46.9 ± 2.2	35.4 ± 2.9	24.5*
Internal capsule	26.1 ± 2.5	18.8 ± 1.2	27.9
Pontine white	27.0 ± 1.1	19.3 ± 2.9	28.5

least in humans, cerebral metabolic rate for O_2 is affected neither by head and body positions [12] nor by natural sleep [10]. It has been previously reported [2] that animal hypnosis may be induced in mesencephalic rabbits, thus implicating a major role of the brain stem reticular formation in the induction and the maintenance of descending inhibitory mechanisms [2, 9]. Other lines of evidence support the hypothesis [18] that the neocortex has a major inhibitory influence on the brain stem center that otherwise produces animal hypnosis. The present results show that the metabolism of the cerebral cortex is greatly affected by animal hypnosis. It may be concluded that, wherever the trigger structures of animal hypnosis are located, their influence produces a diffuse depression of cerebral functional activity.

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